

Qatar-2b

R-1815

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_150429 · created 2026-08-03T04:00:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457141.894625 +/- 8.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.066
Transit depth (Rp/Rs)^2	2.41 +/- 2.05 [%]
Orbital Inclination (inc)	85.62 +/- 1.73
Airmass coefficient 1 (a1)	0.89 +/- 0.33
Airmass coefficient 2 (a2)	0.053 +/- 0.059
Scatter in the residuals of the lightcurve fit is	34.89 %
Best Comparison Star	#2 - [212, 118]
Optimal Aperture	1.93
Optimal Annulus	6.75
Transit Duration (day)	0.0681 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.066 vs 0.16327 (deviation 0.13 σ)
Duration fit vs published	0.0681 d vs 0.0767 d (deviation 1.02 σ)
Mid-time O-C	0.3 min
Depth SNR	2.3
Residual scatter	34.89 %

Light curve

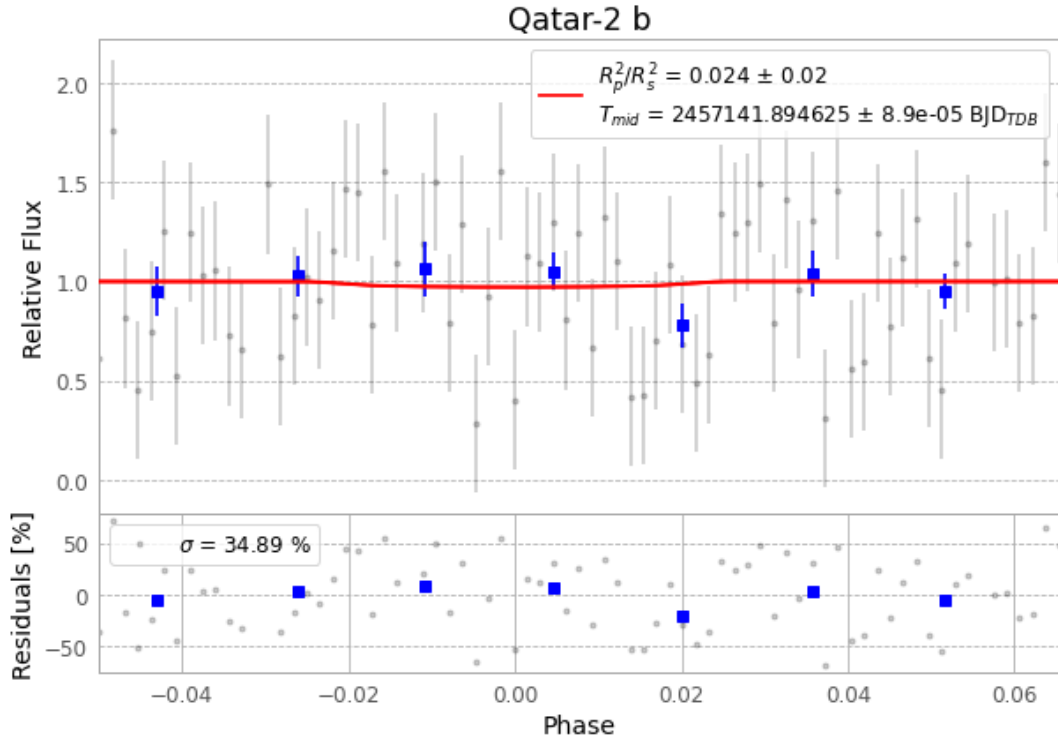


Figure 1 — FinalLightCurve_Qatar-2 b_2015-04-29.png (SHA-256 bf309bb208d5d9ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 431], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a5a946942d713e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar-2150429074212.FITS → Qatar-2150429112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-150429.log (ce3da54e708c...), FinalLightCurve_Qatar-2 b_2015-04-29.png (bf309bb208d5...), FinalLightCurve_Qatar-2 b_2015-04-29.csv (6fcb1d243c77...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:00Z.